

Ian Hutchinson

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EXPERIENCE

Northrop Grumman

Baltimore, Maryland

Principal Software Engineer

June 2022 – Present

- Led development and delivery of a large-scale multi-threaded C++20 application, coordinating engineering efforts across multiple organizations and customer teams.
- Architected and implemented a configurable, containerized software platform leveraging modern C++20, automated testing, GitLab CI/CD, Conan, and Linux-based deployment environments.
- Designed high-performance concurrent software systems emphasizing scalability, reliability, and maintainability in mission-critical environments.
- Modernized a legacy C++ codebase by eliminating technical debt, simplifying dependencies, improving software quality, and reducing long-term maintenance costs.
- Served as Scrum Master and technical lead, mentoring engineers, driving Agile execution, and coordinating cross-functional development efforts.

Software Engineer

October 2019 – May 2022

- Developed high-performance real-time C++ software for mission-critical systems with strict latency, reliability, and throughput requirements.
- Implemented new system capabilities that improved processing efficiency and resource utilization across deployed software platforms.
- Integrated machine learning-based target recognition capabilities into production software systems, enabling automated classification and decision support.
- Collaborated with systems, hardware, and test teams to design, integrate, validate, and deploy complex software solutions.

Associate Software Engineer

January 2019 – September 2019

- Developed embedded software for airborne radar systems supporting surveillance and reconnaissance missions.
- Implemented and tested C++ software components for real-time radar processing applications.
- Supported software integration, troubleshooting, and verification activities throughout the development lifecycle.

Siemens

Atlanta, Georgia

Engineering Leadership Development Program Internship

May 2018 – August 2018

- Developed and commissioned PLC and HMI control systems for industrial manufacturing facilities, supporting automation and production operations.
- Collaborated with customers, technicians, and engineers to deploy and validate automation solutions in production environments.

MSA – The Safety Company

Pittsburgh, Pennsylvania

Electrical Engineering Co-Op

January 2017 – May 2017

- Developed software for automated test fixtures used to validate industrial gas sensors against safety and performance requirements.
- Analyzed test data and collaborated with engineering teams to improve sensor validation and manufacturing processes.

EDUCATION

Georgia Institute of Technology

Atlanta, Georgia

Master of Science in Computer Science

January 2023 – December 2026 (est.)

Penn State University

University Park, Pennsylvania

Bachelor of Science in Electrical and Electronics Engineering

August 2014 – December 2018

TECHNICAL SKILLS

Languages: C++, C, Python, Java, SQL, Bash

Architecture & Systems: Software Architecture, System Design, Distributed Systems, Concurrent Programming, Multithreading, Real-Time Systems, Object-Oriented Design

Development & DevOps: Linux, Git, GitLab CI/CD, Docker, Conan, Automated Testing, Continuous Integration, Agile/Scrum

Databases & Libraries: SQL, SQLite, Eigen, STL